KENDALL ALEXANDER REID

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RESEARCH INTERESTS

I am interested in creating AI systems for scientific research in human genetics and genomics. Current models excel on well-defined problems but struggle when evidence is uncertain or contradictory. I aim to build systems that overcome these limitations, and to develop rigorous evaluations to ensure that their reasoning is transparent and conclusions can be trusted. By harnessing these models for human genetics, we can discover the mechanisms of disease and identify therapeutic targets.

EDUCATION

Johns Hopkins University — Baltimore, MD
Ph.D., Computer Science. Advisors: Alexis Battle and Brian Caffo — September 2026 – expected 2030

Johns Hopkins University — Baltimore, MD
M.S., Bioinformatics — September 2024 – present

University of Maryland — College Park, MD
Post-baccalaureate, Biology and Chemistry — 2021 – 2024

Rutgers University — New Brunswick, NJ
B.A., Mathematics and Economics, *summa cum laude* — 2013 – 2017
Thesis: *A High-Frequency Methodology for Studying Sector Volatility Transmission*

PUBLICATIONS

Peer-Reviewed Journal Articles

1. S. Sarma, H. Truscott, D. Xu, **K. A. Reid**, L.-S. Chu, J. Chen, J. J. Gray. "Can we extract physics-like energies from generative protein diffusion models?" *PRX Life* (American Physical Society), accepted, in press. Preprint: doi:10.1101/2025.11.28.690021
2. P. Kasaie, Y. Zhao, E. Humes, L. Gerace, **K. A. Reid**, et al. "Impact of an early body mass index maintenance intervention at ART initiation on diabetes risk: insights from a simulation study." *Annals of Epidemiology*, accepted, in press. Preprint: doi:10.1101/2025.11.05.25339607
3. L. L. Tosi, E. N. Rajah, A. P. Gillies, M. Kim, **K. A. Reid**, R. I. Gafni. "Burosumab treatment in an adult with FGF23-mediated hypophosphatemia due to cutaneous skeletal hypophosphatemia syndrome." *Journal of the Endocrine Society* 10(4): bvag040, 2026. doi:10.1210/jendso/bvag040
4. L. L. Tosi, K. Templeton, A. M. Pennington, **K. A. Reid**, B. D. Boyan. "Influence of sex and gender on musculoskeletal conditions and how they are reported." *Journal of Bone and Joint Surgery (American Volume)* 106(16): 1512–1519, 2024. doi:10.2106/JBJS.24.00194

Manuscripts Under Review

1. **K. A. Reid**, F. Goes. "A blended genome-exome polygenic risk scoring of ketamine response in patients with drug-resistant depression." Under review, *JAMA Psychiatry*, 2026.

Preprints

1. **K. A. Reid**, G. Stein-O'Brien, and E. Guven, "Topology Matters: The Trade-off Between Wasserstein Critics and Discriminators in Single-Cell Data Integration," bioRxiv, Nov. 13, 2025 (version 3, May 5, 2026). doi:10.1101/2025.11.12.688061.
2. P. S. L. Schäfer, **K. A. Reid**, Z. Boldyga, E. D. Aksu, H. Hakem, and J. Saez-Rodriguez, "Towards principled evaluation of single-cell perturbation prediction models," bioRxiv, Jul. 27, 2026. doi: 10.64898/2026.07.23.740433.

RESEARCH EXPERIENCE

Johns Hopkins University, Biostatistics, and Neuroscience Baltimore, MD
Senior Programmer Analyst *September 2025 – September 2026*

- Designed and led a study of whether latent variable models isolate known gene programs in single-cell data. First author; manuscript in preparation.
- Led the computational biology for a companion study applying latent variable models to amyotrophic lateral sclerosis (ALS) and frontotemporal dementia (FTD) to identify similarities and differences in disease pathology. Second author; manuscript in preparation.

Johns Hopkins University, Department of Epidemiology Baltimore, MD
Programmer Analyst *June 2024 – September 2025*

- Contributed to an agent-based simulation library for HIV population dynamics using Dask.
- Co-authored a study that used this library to forecast the effect of a BMI maintenance intervention on diabetes risk.
- Led the interpretability analysis for a study of virologic failure in people with HIV, investigating whether a linear Cox model and a nonlinear deep survival network identified the same influential risk factors. Third author; manuscript in preparation.

Johns Hopkins University Baltimore, MD
M.S. Student, Bioinformatics *September 2024 – present*

- Recruited a team of researchers from across the globe to investigate the discordance in metrics used to evaluate perturbation prediction models, resulting in a preprint. Second author; submission in progress.
- Designed a controlled comparison of Wasserstein critics and discriminators to study the tradeoff between removing batch effects and preserving biology in adversarial batch correction for single-cell data. First author; manuscript in preparation.
- Identified that computing energies along a protein diffusion model's reverse diffusion trajectory with trapezoidal integration is a discrete realization of the Stratonovich integral, contributing to a publication on protein energies in diffusion models.
- Built a blended genome-exome polygenic risk score for ketamine response in treatment-resistant depression, in collaboration with the Broad Institute. First author; under review.

Children's National Hospital Washington, DC
Research Assistant, Orthopedic Surgery *May 2022 – June 2024*

- Directed longitudinal data collection for an industry-sponsored clinical trial in a rare skeletal dysplasia.

INDUSTRY EXPERIENCE

Productivity Industries San Francisco, CA
Co-Founder; Data Scientist, Computer Vision and Natural Language Processing *January 2019 – January 2021*

- Built the company's document-parsing system: multimodal models combining computer vision and natural language processing to predict the reading order of unstructured text in PDF documents.
- Built the continuous-integration infrastructure for training, evaluating, and deploying the models.

Iunu Seattle, WA
Data Scientist, Computer Vision and Artificial Intelligence *June 2017 – July 2018*

- Built AI-based control systems for greenhouse temperature, water and light.
- Developed a deep learning pipeline that measured plant material in multispectral images and predicted yield from image time series.
- Trained convolutional networks to identify plant disease and alert growers prior to plant loss.

PRESENTATIONS

1. **K. A. Reid**, M. Brooks, Y. Zhao, E. Humes, Y. Xu, K. N. Althoff. "Do machine learning and traditional regression models identify the same influential risk factors? A case study of virologic failure in people with HIV." 9th Annual Johns Hopkins Research Symposium on Engineering in Healthcare, Baltimore, MD, December 2025. **Best Poster Award.**
2. H. Giles, **K. A. Reid**, G. Stein-O'Brien. "Biological disentanglement of ALS and FTD." 26th Annual Packard Center ALS Research Symposium, Baltimore, MD, March 2026.
3. K. N. Althoff, P. Kasaie, C. Lesko, E. Humes, **K. A. Reid**, et al. "Weight maintenance at ART initiation and impact on subsequent diabetes risk: triangulating inferences from two analytic approaches." ACTG Annual Network Meeting, Washington, DC, June 2026.

TEACHING AND MENTORING

- Co-mentored an M.S. Biostatistics thesis on generative modeling for single-cell data analysis (with B. Caffo); the student was subsequently admitted to a Ph.D. program at the University of Chicago.
- Mentored medical students in computational analysis of SNP data, leading to national conference presentations. Children's National Hospital, 2022–2024.

HONORS AND AWARDS

- Best Poster Award, Johns Hopkins Research Symposium on Engineering in Healthcare, 2025
- Henry Rutgers Scholar Award, for outstanding research paper, Rutgers University, 2017
- John C. Daniels Award, for outstanding achievement in economics, Rutgers University, 2017
- Milton Friedman Distinguished Scholar, Rutgers University, 2017

TECHNICAL SKILLS

Languages Python, SQL

Machine learning PyTorch, PyTorch Lightning, TensorFlow, Hugging Face, scikit-learn, OpenCV

Scientific computing NumPy, pandas, Dask, scverse, matplotlib, seaborn, VCFtools

Infrastructure Linux, Git, Docker, SLURM, Snakemake, Nextflow, GitHub Actions, pytest